

Edulytix

Transforming Educational Data into Actionable Intelligence

Project Description

Edulytix is a next-generation **data analytics and decision intelligence platform** designed to help educational institutions make informed, evidence-based decisions in admissions, marketing, and enrollment management.

Built specifically for the **Mays Business School's FLEX MBA Program**, Edulytix aims to bridge the gap between **raw institutional data** and **strategic decision-making** by providing a unified platform that integrates three key capabilities — **data visualization, predictive analytics, and AI-driven insights**.

In today's competitive educational landscape, universities must efficiently allocate their marketing budgets, anticipate student enrollment trends, and evaluate the effectiveness of their promotional efforts. However, decision-makers often face challenges such as fragmented data systems, inconsistent reporting formats, and limited analytical visibility. **Edulytix** solves this problem by providing a holistic and interactive solution that not only visualizes data trends but also interprets them through intelligent forecasting and conversational analytics.

By leveraging historical admissions and marketing data, the platform will help academic leaders answer key strategic questions such as:

- *What promotional activities generate the most applications?*
- *How much marketing budget should be allocated next semester?*
- *What enrollment volume can we forecast for upcoming terms?*

Strategic Impact

- **For Decision Makers:** Edulytix provides an executive-level overview of critical recruitment and marketing indicators, empowering directors and deans to make data-backed budgetary and operational decisions.
- **For Analysts & Administrators:** It simplifies the analytical workflow by automating data cleaning, visualization, and forecasting tasks, reducing manual reporting efforts.
- **For Institutional Growth:** By quantifying marketing effectiveness and predicting future enrollments, Edulytix helps optimize spending, increase recruitment efficiency, and improve the overall student acquisition strategy.



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Project Scope

1. Data Visualization Dashboard

- **Goal:** Create an interactive Power BI dashboard (or equivalent BI tool) that visualizes key performance metrics from existing admissions and marketing datasets.
 - **Features:**
 - Display historical application trends, conversion rates, and demographic patterns.
 - Correlate marketing expenditures (ads, events, digital campaigns) with enrollment outcomes.
 - Provide drill-down capabilities for program-specific or seasonal insights.
 - **Outcome:** Enables senior administrators to evaluate the ROI of past marketing activities and optimize future spending decisions.
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2. Predictive Forecasting Engine

- **Goal:** Develop a forecasting model to predict future enrollment numbers and seasonal trends using machine-learning techniques.
 - **Features:**

Use historical data to forecast applications for upcoming months or semesters.

 - Allow users to input custom forecast horizons (e.g., 1, 3, or 6 months).
 - Integrate time-series or regression models (e.g., Linear Regression, ARIMA, or Prophet).
 - **Outcome:** Provides a data-backed prediction of expected application volumes, assisting leadership in resource planning and admissions strategy.
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3. AI Chatbot Integration

- **Goal:** Integrate a GPT-powered conversational assistant capable of answering predefined data-driven queries.
- **Features:**
 - Natural-language interaction with the database (e.g., *"What was our total marketing spend last year?"*).
 - 10–15 pre-programmed key queries covering admissions, budget allocation, and forecast summaries.
 - Extendable chatbot framework for future custom queries.

- **Outcome:** Streamlines decision-making by allowing quick, on-demand access to institutional data insights.

Project Timeline

Phase	Timeline	Description & Deliverables
Phase 1: Data Preparation & Planning	Month 1	Collect existing datasets, define metrics, clean and preprocess data. Finalize database schema and data pipelines.
Phase 2: Dashboard Development	Months 2–3	Build Power BI dashboards with relevant KPIs, visualizations, and user access controls. Deliver initial prototype for feedback.
Phase 3: Predictive Modeling	Month 4	Implement forecasting models; validate accuracy with historical data. Integrate predictive outputs into dashboard.
Phase 4: AI Chatbot Development	Month 5	Develop and connect the GPT-based chatbot to the backend. Configure 10–15 query templates and perform user testing.
Phase 5: Integration, Testing & Deployment	Month 6	Conduct full system testing, refine based on feedback, and prepare a final working demo for stakeholders.

Future Vision

After successful implementation for the **FLEX MBA Program**, Edulytix can be expanded into a **scalable SaaS platform** for other departments and universities. The next phase will explore **monetization strategies**, such as licensing, customization for institutions, or integration with learning management systems.

Summary Statement

Edulytix bridges the gap between data and decision-making in higher education. By combining powerful visual analytics, predictive intelligence, and conversational AI, it equips academic institutions with the tools they need to **forecast smarter, spend wiser, and manage admissions more effectively**.